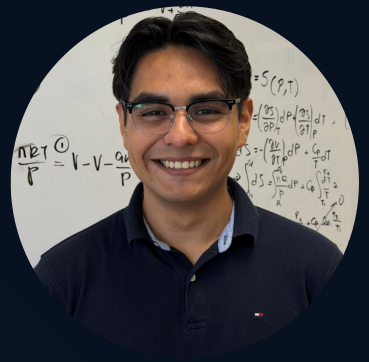
EvoEmu

In silico directed evolution for protein conformations



The Team!



Diego Velazquez
Nanoengineering B.S.
Tijuana Technological Institute



Ibrahim Aasim
Computer Science B.S.
UC San Diego



Nikhila Manikandan
Computer Science B.S.
Texas A&M



Gilbert Shen
Bioengineering B.S.
UC Berkeley



Antranig Baghdassarian
Biomedical Engineering B.S.
Cornell University

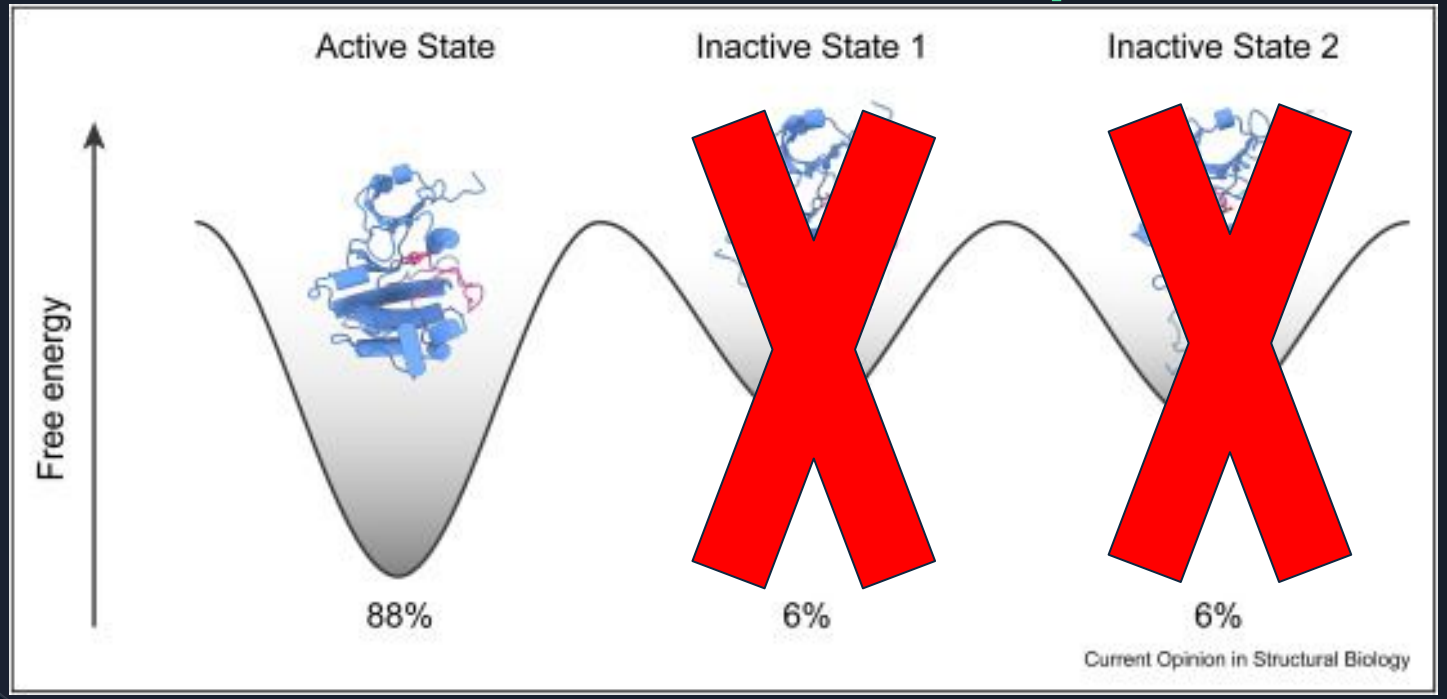


Imane Jennane
Robotics M.S.C
EPFL - Stanford

Proteins are Key to Life

- 3D structure dictates protein function → Misfolded Proteins → Various Diseases
- Proteins are dynamic → Multiple conformations → Conformational Landscape
- Alpha Fold & early protein studies only consider one conformation:

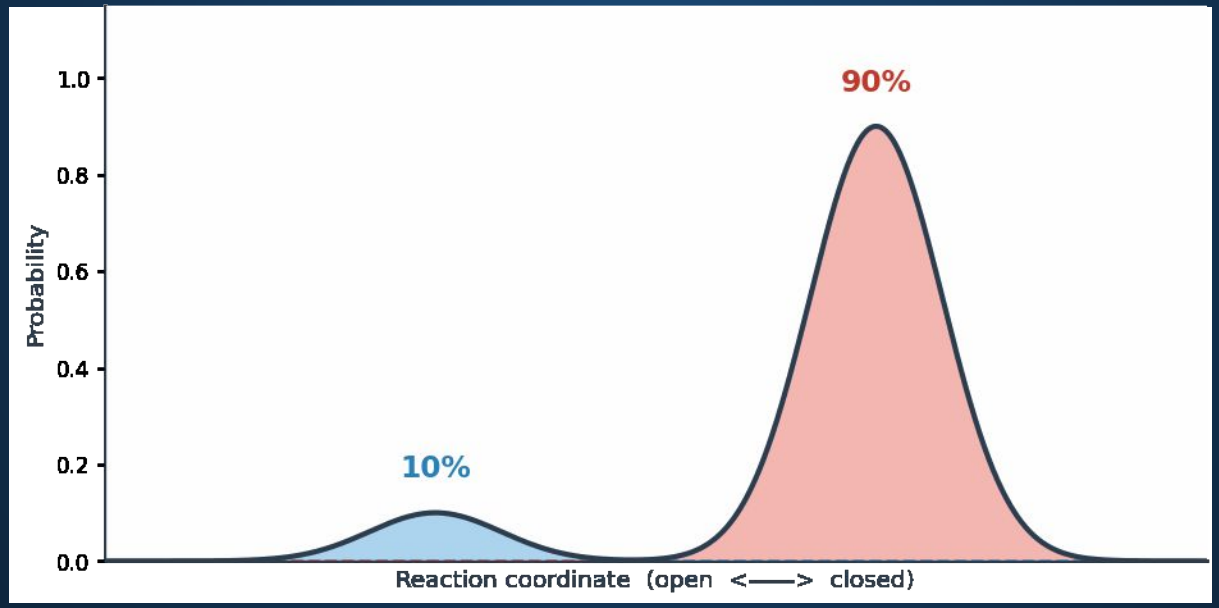
Conformational Landscape



In silico directed evolution

Modulating the conformational landscape of a protein

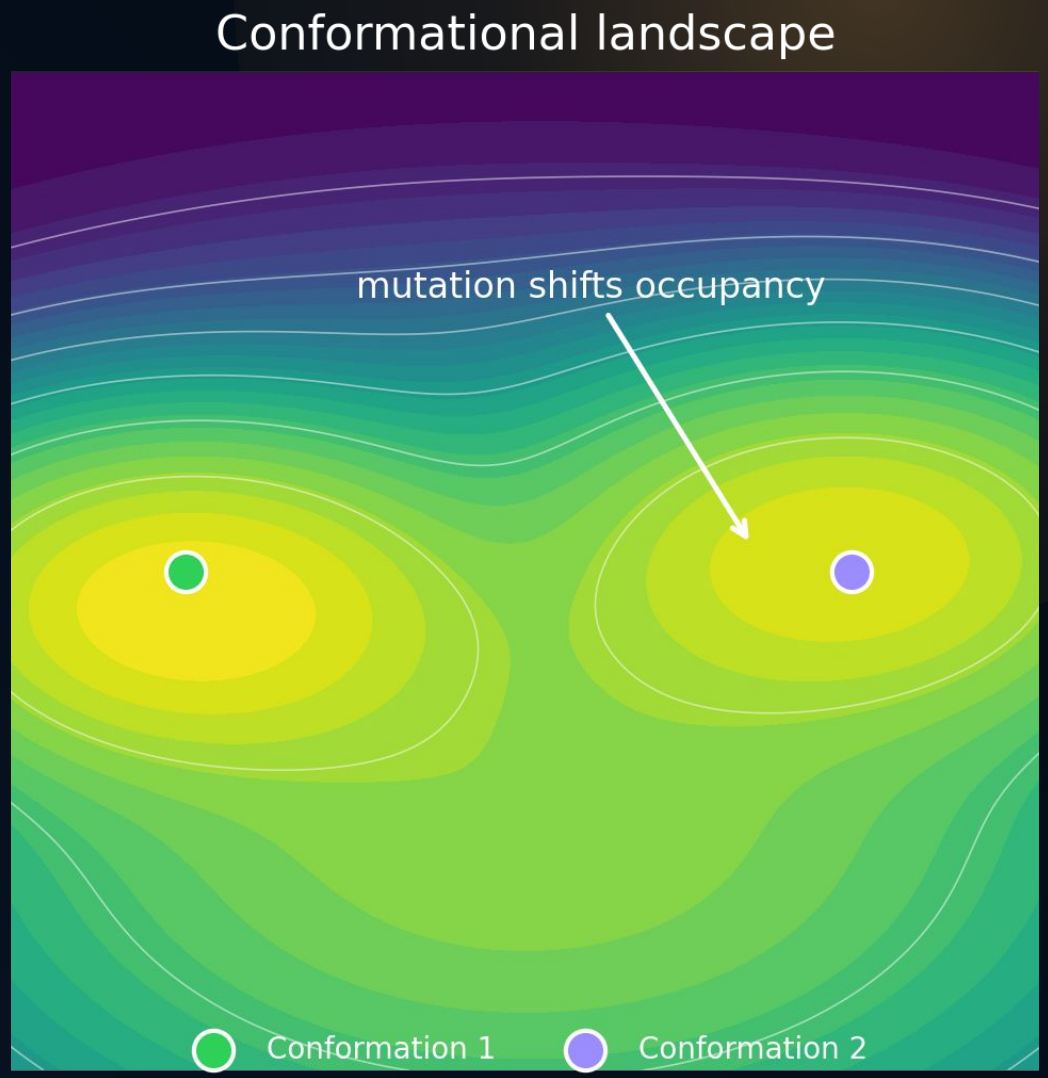
Drugs can affect the conformational landscape, but usually they just lock it to one position.
Determining the allosteric effect is expensive and time consuming.
EvoEmu engineers sequences for them to fit a user-defined conformational landscape.



We propose LLR as a metric to compare the relative abundances of two states. LLR = $\ln(P_2/P_1)$

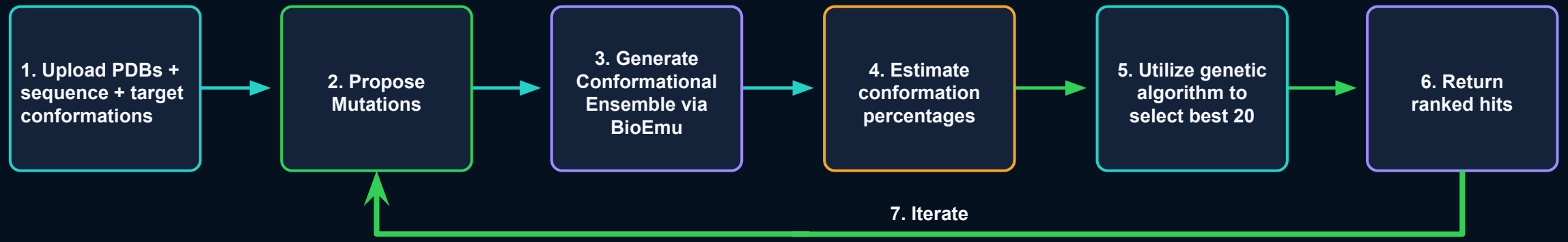
EvoEmu As a Solution

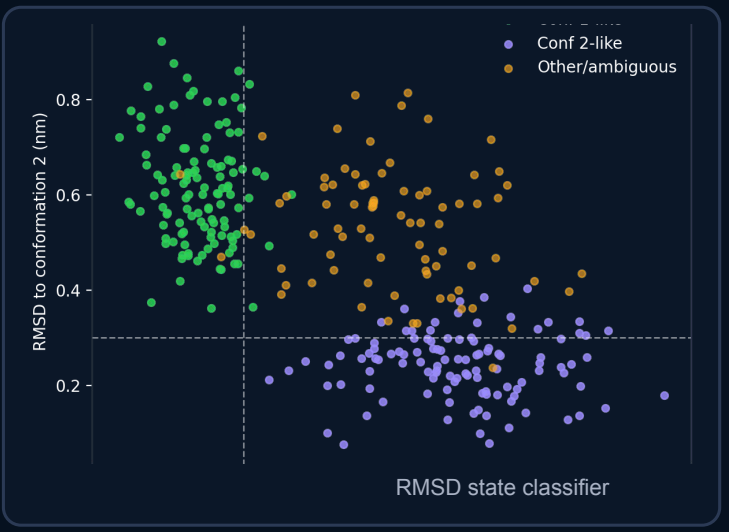
- BioEMU predicts the conformational landscape of a given protein sequence. We compute LLR and use it as score for directed evolution to:
- Fit to LLR
  - Generates mutations to shift conformational landscape towards goal
- Fit to structure
  - Upload desired structure and EvoEmu will mutate a starting sequence to stabilize desired structure

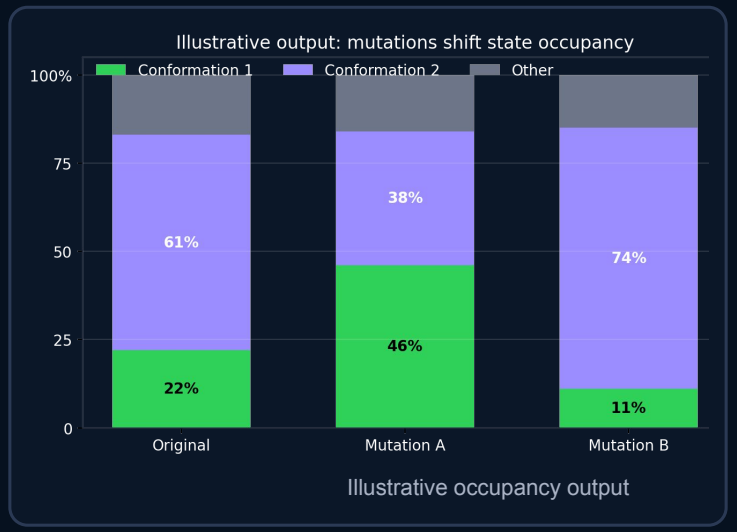


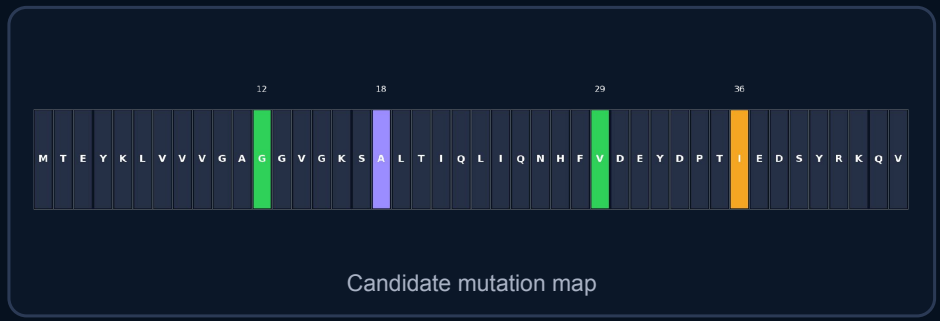
End-to-End Workflow

upload → propose → sample → score → rank → repeat.









Core metrics: RMSD-to-state and ensemble occupancy

EvoEmu Genetic Algorithm Search

Iteratively refine around the most promising variants instead of exhaustively testing the full mutation space

Initial sequence → Generate 100 → Score by LLR → Keep top 20 → Next 100 parents

5 rounds × 100 candidates = 500 total sequences



dim = low LLR / far from goal

bright = high LLR / goal-like

best candidates become parents

Why this is practical: instead of waiting days for exhaustive search, EvoEmu/ESM-2 explores 500 focused variants by repeatedly keeping the top 20 and sampling nearby sequence space.

EvoEmu

Engineer protein sequences toward a target conformation

In silico directed evolution. ESM-2 proposes mutations, BioEmu scores each candidate's conformational landscape, and a genetic search evolves the sequence toward your target.

ESM-2	BioEmu	Top-20
guided mutation	conformational scoring	elite per round



CONFIGURATION

Starting sequence — single-letter amino acid codes

```
TTYKYILNLKFAFGDTNSEAVDAAEAEKKFKQYANDHGVDGEWTYDDATKTFTVTAKD
SHADRVRELAQRLRQRPRVERVEITEVTE
```

Optimization mode

Fit to LLR | Fit to Structure

Auto detect states
Fills both viewers from the input sequence via ESMFold

Impact

- ➢ Cuts months of lab testing down to a single computational run
- ➢ Flags mutations worth testing for drug discovery before any lab work
- ➢ Shows how disease mutations shift protein behavior, not just shape
- ➢ Gives engineers a way to push proteins toward the state they want

Impact

Fast Protein Engineering Candidate Search:

KRAS Cancer

➢ Mutant KRAS stays stuck "on" → cells grow out of control
➢ Drugs just lock it "off" instead
➢ EvoEmu restores the normal on/off switch

RuBisCO & Crop Yield

➢ Plants lose 30% of CO_2 capture to a wasteful side reaction
➢ Two modes: useful vs. wasteful
➢ EvoEmu shifts the balance toward useful → more food

Therapeutic Antibodies

➢ Antibodies spend time in non-binding conformations → wasted dose
➢ EvoEmu favors the "ready to bind" shape
➢ Same drug, stronger effect, lower dose

Basic Research Tool:

Easily stabilize a desired conformation for study with tools such as NMR, etc
Protein engineering with most of the native sequence intact (as opposed to ProteinMPNN)

Future Directions

Where EvoEmu goes next

01

Increase Robustness

- Test across diverse protein families
- Benchmark LLR stability vs sample size
- Validate genetic algorithm convergence
- Add confidence intervals to LLR scores

02

More Input Modalities

- Structure-to-sequence: input target PDB directly
- Multi-state targets beyond 2 conformations
- Ligand-aware inputs (post-BioEmu update)
- ProteinMPNN · Rosetta · phylogenetic MSA proposals

03

Experimental Validation

- Synthesize top 5 hits, express in E. coli
- Validate via smFRET and NMR
- Compare experimental ratios to LLR predictions
- Close loop: fine-tune BioEmu via PPFT

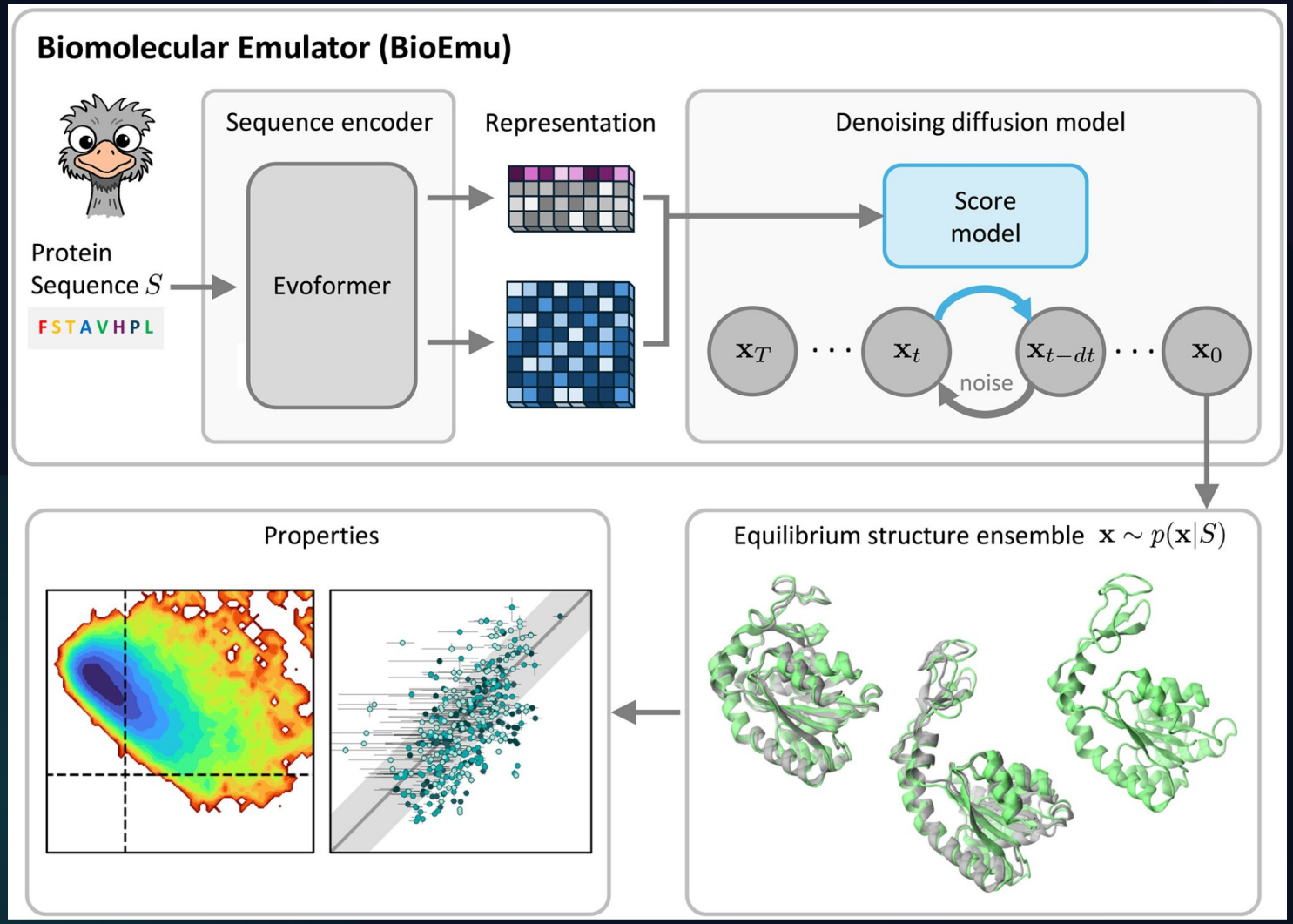
The roadmap: more robust · more flexible · experimentally grounded

Supplemental Slides

1. BioEmu Functionality
2. Scoring
3. Mutation Generation
4. Cost

Thanks for listening!

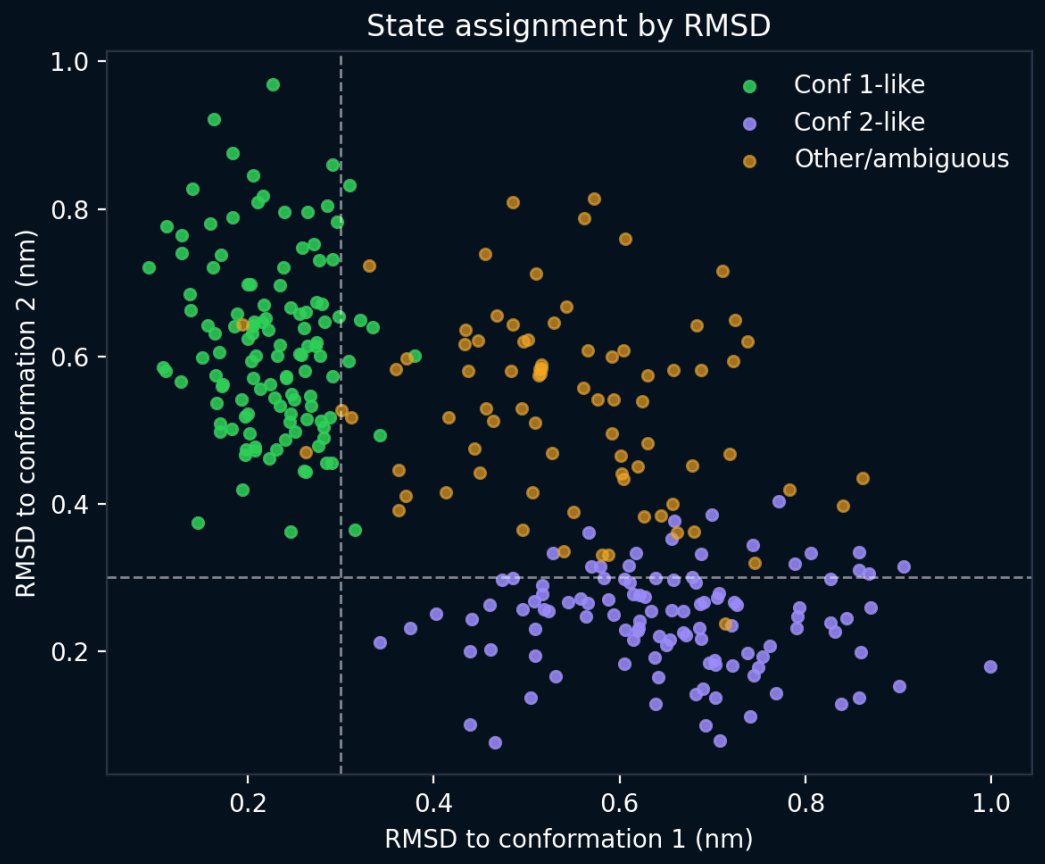
BioEmu Model and Workflow



https://www.science.org/doi/10.1126/science.adv9817

Scoring: every sampled structure votes for a state

EvoEmu compares each BioEmu sample to both reference PDBs.



Classification rule

For each BioEmu sample:

r1 = RMSD(sample, conformation1.pdb)
r2 = RMSD(sample, conformation2.pdb)

conf1-like if r1 < r2 and r1 < threshold
conf2-like if r2 < r1 and r2 < threshold
otherwise: other

Occupancy estimate

P(conf1) = n_conf1 / n_samples
P(conf2) = n_conf2 / n_samples
P(other) = n_other / n_samples

Wilson confidence intervals are reported for probabilities.

Scoring code: evoemu/score_states.py

Predicted candidates; not experimental validation.

Mutation generation: conservative exploration first

EvoEmu searches sequence space without brute-forcing the entire combinatorial universe.



Single mutation

A49I means:
wild type A at position 49 becomes I

The code chooses random positions inside the mutable window, e.g. 30–70, then chooses a replacement amino acid.

Conservative groups (ESM like)

A,V,L,I,M → hydrophobic-ish
F,W,Y → aromatic
S,T,N,Q → polar
K,R,H → positive
D,E → negative

Multi-mutants: current setup samples 2–3 mutations per candidate, e.g. A49I+P65A or N30S+A49I+L70V.

Mutation code: evoemu/mutate_sequence.py

Predicted candidates; not experimental validation.

Cost

Run	Residue search space	BioEmu evaluations / iterations	Requested samples	Scored samples after BioEmu filtering	Time on H100
Multi-mutant local window	residues 30–70, 41 residues	55 evaluations	4,864 samples	4,593 samples	3,641 sec
Multi-mutant all residues	residues 1–90, 90 residues	55 evaluations	4,864 samples	4,545 samples	3,743 sec

1 hr/5,000 samples * $3.29/hr = $3.29/5,000 samples

